

COVID-19 — Comprehensive Clinical Guide

1. Background and Epidemiology

COVID-19 (Coronavirus Disease 2019) is caused by SARS-CoV-2, a novel betacoronavirus first identified in Wuhan, China in December 2019. The WHO declared COVID-19 a pandemic on March 11, 2020. By early 2023, over 700 million confirmed cases and 7 million deaths had been reported globally.

India experienced two major devastating waves — Wave 2 (April-June 2021) was characterized by the Delta variant (B.1.617.2), causing unprecedented mortality and healthcare system strain. The Omicron variant (B.1.1.529) and its subvariants dominated globally from late 2021, causing more transmissible but generally less severe disease.

SARS-CoV-2 continues to evolve. XBB.1.5, JN.1, and subsequent subvariants of Omicron have shown progressive immune evasion. Ongoing surveillance and updated vaccines are essential parts of pandemic preparedness.

2. Virology and Transmission

SARS-CoV-2 is a positive-sense single-stranded RNA virus. Its spike (S) protein binds to ACE2 receptors on human cells (lungs, gut, heart, kidneys, brain). The receptor-binding domain (RBD) of S protein is the primary target of neutralizing antibodies and most COVID-19 vaccines.

Transmission routes: primarily respiratory — large droplets (>5 micrometers, short range) and fine aerosols (<5 micrometers, airborne). Aerosol transmission is dominant indoors in poorly ventilated spaces. Contact/fomite transmission is possible but plays a minor role.

Incubation period: 1-14 days (mean 4-6 days for Omicron variants). Pre-symptomatic and asymptomatic transmission is common. Infectious period: 1-2 days before symptom onset to 5-10 days after (longer in immunocompromised). Omicron has shorter serial interval (~2-3 days) than Delta, facilitating faster spread.

3. Clinical Spectrum

Approximately 80% of COVID-19 cases are mild or asymptomatic. 15% develop severe disease requiring oxygen. 5% become critically ill (ICU, mechanical ventilation). Overall infection fatality rate varies by variant, vaccination status, and age.

Common symptoms: fever (77%), dry cough (67%), fatigue (38%), loss of smell and taste (anosmia/ageusia — characteristic of earlier variants, less common with Omicron), headache, myalgia, sore throat, rhinorrhoea, gastrointestinal symptoms (nausea, diarrhoea).

Severe disease: hypoxemia (SpO₂ <94%), bilateral interstitial infiltrates (COVID pneumonia), cytokine storm (hyperinflammation), ARDS (acute respiratory distress syndrome), multi-organ failure.

High-risk groups: age >60, male sex, diabetes, hypertension, cardiovascular disease, chronic lung disease, obesity (BMI >30), immunocompromised state (HIV, organ transplant, cancer on chemotherapy), pregnancy.

4. Diagnosis

	Type	Window / Notes
	Molecular — gold standard	Days 1-14; nasopharyngeal/oropharyngeal swab
st (RAT)	Antigen detection	Days 1-7; lower sensitivity than PCR
	Antigen detection	Easy, same-day; confirm negative in symptomatic
M)	Serology	Positive after Day 7-14; not for acute Dx
	Imaging	CO-RADS/CORADS-6; GGOs, crazy paving patte
D-dimer, Ferritin	Severity markers	Elevated in severe disease
	Oxygenation	Critical for triage and ventilation decisions

5. Treatment — Mild to Moderate

Mild disease (no hypoxemia, no high-risk features): symptomatic treatment at home.

Paracetamol 500-1000 mg every 6-8 hours for fever and myalgia. Adequate hydration (2-3 litres/day). Rest and self-monitoring (SpO₂ ≥94% target).

Antivirals for high-risk mild/moderate patients (within 5 days of symptom onset):

nirmatrelvir-ritonavir (Paxlovid, Pfizer) 300/100 mg BD for 5 days — reduces hospitalization by 89% in unvaccinated high-risk adults. Multiple drug interactions (CYP3A). Contraindicated

in severe renal impairment.

Molnupiravir (Lagevrio, Merck): 800 mg BD for 5 days. Modest 30% reduction in hospitalization. NOT recommended in pregnancy (teratogenic potential).

Remdesivir (Veklury): IV antiviral, approved for hospitalized patients requiring supplemental oxygen but NOT mechanical ventilation. Reduces time to recovery. WHO Solidarity trial showed equivocal mortality benefit.

6. Treatment — Severe and Critical

Supplemental oxygen: the cornerstone of management. Standard nasal cannula → high-flow nasal oxygen (HFNO) → non-invasive ventilation (NIV: CPAP/BiPAP) → mechanical ventilation. Proning (prone positioning) for 12-16 hours/day improves oxygenation in moderate-severe COVID pneumonia.

Dexamethasone 6 mg daily for 10 days: first drug proven to reduce mortality in severe COVID (RECOVERY trial — 35% relative reduction in deaths in ventilated patients, 23% in those on oxygen). Only beneficial in those requiring oxygen. Harmful if given to non-hypoxic patients.

Anti-IL-6 agents: tocilizumab (8 mg/kg IV single dose) for patients with rapidly increasing O₂ requirements and systemic inflammation (CRP >75 mg/L). Combined with dexamethasone reduces 28-day mortality.

Baricitinib (Janus kinase inhibitor, 4 mg daily): alternative to tocilizumab with comparable mortality benefit in severe COVID. Also approved for COVID-19.

Anticoagulation: prophylactic dose LMWH (enoxaparin 40 mg SC daily) for all hospitalized COVID patients unless bleeding contraindication. Therapeutic anticoagulation for confirmed VTE or LVO stroke during COVID.

7. COVID-19 Vaccines

	Doses	India Availability
d virus	2 doses	Yes (Serum Ins)
	2 doses	Yes (Bharat Bi)
	2 doses	Yes (Biological)
	2 + booster	Limited access

	2 + booster	Limited access
	2 doses	Approved in India

All approved COVID-19 vaccines have shown high efficacy (70-95%) against severe disease and death in clinical trials. With emergence of Omicron variants, vaccine effectiveness against infection has waned but protection against severe disease/hospitalization largely maintained (especially with booster doses).

Booster vaccines: bivalent boosters (targeting original strain + Omicron BA.4/BA.5) shown to improve immune response against Omicron subvariants. India's National COVID-19 Vaccination Programme began January 16, 2021 — achieved 2 billion doses administered by July 2022.

8. Long COVID

Long COVID (Post-COVID-19 Condition) is defined by the WHO as symptoms persisting or beginning after ≥ 3 months from the onset of confirmed COVID-19, lasting at least 2 months, and not explained by an alternative diagnosis.

Prevalence: approximately 10-20% of COVID-19 cases develop Long COVID. Estimated 65 million people globally affected.

Common symptoms: post-exertional malaise (PEM) — worsening of symptoms after physical or cognitive effort, profound fatigue, cognitive impairment/brain fog (difficulty concentrating, memory problems), dyspnoea, chest pain/tightness, joint/muscle pain, palpitations, sleep disturbances, depression and anxiety, anosmia.

Mechanisms proposed: viral persistence, immune dysregulation (autoantibodies), microbiome disruption, endothelial dysfunction and microthrombi, reactivation of latent viruses (EBV, CMV), mitochondrial dysfunction.

Management: multidisciplinary, symptom-based. Pacing (activity management) to prevent PEM. Pulmonary rehabilitation for breathlessness. Cognitive rehabilitation for brain fog. Psychological support. SSRI may help neurological symptoms. No proven curative treatment yet.

9. Infection Prevention and Control

Hand hygiene: frequent handwashing with soap and water for 20 seconds, or hand sanitizer (60-70% alcohol). Particularly before eating, after touching surfaces, after coughing/sneezing.

Masks: N95/KN95 respirators provide highest protection against aerosol transmission. Surgical masks reduce outward spread. Cloth masks have limited efficacy against aerosol transmission.

Ventilation: increasing outdoor air ventilation (open windows, fans), HEPA air purifiers, and CO2 monitoring (>1000 ppm indicates poor ventilation) are key infection control tools.

Isolation: COVID-19 positive individuals should isolate for at least 5 days (CDC guidelines) or longer if symptomatic beyond Day 5. Wear a well-fitting mask when around others for 10 days. In India, home isolation guidelines issued by MoHFW should be followed.

10. COVID-19 in India — Specific Considerations

India's healthcare response: establishment of 1L+ COVID-dedicated beds, Ayushman Bharat digital health ID integration, telemedicine growth, ICMR COVID testing scale-up from 100 to 2 million tests/day at peak. Challenges: healthcare infrastructure overload during Wave 2, oxygen shortage crisis.

Mucormycosis (Black Fungus): rare but life-threatening fungal infection affecting sinuses, brain, and lungs. COVID-associated mucormycosis (CAM) emerged significantly in India during Wave 2, associated with steroid overuse in non-hypoxic patients and uncontrolled diabetes. Treatment: IV amphotericin B + surgical debridement.

Children and MIS-C: most children have mild COVID-19. Multisystem Inflammatory Syndrome in Children (MIS-C) is a rare but severe post-COVID complication — features of Kawasaki disease, shock, cardiac involvement. Treated with IVIG, aspirin, and corticosteroids.

COVID and tuberculosis: COVID-19 disrupted TB services globally, with India seeing an estimated additional 200,000+ TB deaths due to disrupted diagnosis and treatment. Post-COVID pulmonary fibrosis can overlap with TB sequelae radiologically. Dual COVID/TB infections reported.